

NATIONAL INSTITUTE OF TECHNOLOGY, WARANGAL

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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***Abstract*—Imagined-word recognition from electroencephalography (EEG) represents a foundational brain–computer interface (BCI) challenge because the neural patterns underlying covert linguistic articulation are weak, non-stationary, and highly variable across sessions and individuals. This paper presents an advanced Multi-Scale Dual-Geometry Riemannian Covariance with Adaptive Genetic Optimization (MDRC-AGA) architecture that bridges the gap between binary proof-of-concept baselines and practical multi-class BCI communication. The starting point is a rigorous reproduction of the primary reference paper by Lara-Arellano et al. (*Eng*, MDPI, 2025), which achieved 96.0% accuracy on binary *Yes* vs *No* (924 real trials) using a 228-bit Genetic Algorithm (GA) with *k*-Nearest Neighbors (*k*NN). However, the primary paper could not evaluate multi-class decoding across all nine vocabulary classes due to the combinatorial computational cost of evaluating 228-bit genomes across 10 classifiers. Furthermore, conventional multi-class EEG pipelines that rely on flat statistical and spectral moments fail to scale to larger vocabularies because univariate scalar moments treat electrodes as isolated sensors and destroy inter-channel phase synchronization.**

To overcome these limitations, our proposed MDRC-AGA maps multichannel EEG signals into the Riemannian manifold of Symmetric Positive Definite matrices S_{++}^{n}, engineering a 1,695-dimensional multi-scale filter bank spanning 6 canonical bands (630D), 7 fine causal sub-bands from 4 to 45 Hz (735D), and a speech-localized 10-channel sub-montage (330D). An evolutionary MultiScaleBlockAGA optimizes continuous band-weight projections while strictly preserving Riemannian positive-definiteness, and a Calibrated Stacker Ensemble (L2-Logistic Regression, Shrinkage-LDA, Platt-Ridge, and Soft Probability Voting) yields well-calibrated posterior distributions. Evaluated across the complete 9-class Mendeley EEGIS corpus (4,044 real trials) on a held-out 20% test partition (809 real trials), our framework achieves 90.85% test accuracy (735/809 correct) with a real-time CPU latency of 11.2 ms per epoch, outperforming the un-optimized multi-class baseline by 20.46 percentage points while demonstrating practical clinical viability.

***Index Terms*—Brain–computer interface (BCI), covert imagined speech, EEG signal processing, Riemannian manifold geometry, Symmetric Positive Definite (SPD) covariance, MultiScaleBlockAGA, probability calibration, assistive neuro-prosthetics.**

I. INTRODUCTION

Brain–computer interfaces (BCIs) aim to infer a user's communicative and motor intent directly from electrophysiological brain dynamics, translating neural activity into control commands or communication channels without requiring peripheral neuromuscular engagement. Electroencephalography (EEG) represents the dominant non-invasive modality in BCI research due to its portability, safety, high temporal resolution at the millisecond scale, and accessible hardware cost [1]. EEG-based communicative neuro-prostheses hold profound therapeutic potential for individuals with severe motor disabilities, including locked-in syndrome (LIS), amyotrophic lateral sclerosis (ALS), and brainstem stroke [2].

Among BCI paradigms, imagined or covert speech recognition —the internal vocalization of words without acoustic phonation or articulatory muscle movement—represents one of the most natural yet difficult decoding challenges [3]. Unlike overt speech recognition, where rich acoustic waveforms provide distinct spectral and phonological landmarks, imagined speech must be decoded solely from microvolt-level potential fluctuations recorded at the scalp. These electrical patterns suffer from severe signal-to-noise ratio (SNR) degradation, non-stationarity, volume conduction through the skull, and cross-session covariate shift [4].

The task addressed in this work is nine-class imagined-word classification using the public Mendeley EEGIS corpus [5] (DOI: 10.17632/73g4fw884c.1), comprising 4,044 real trials across nine essential communicative prompts: *Rest, Help (Ayuda), Bathroom (Baño), Pain (Dolor), Thanks (Gracias), Hungry (Hambre), No, Thirsty (Sed), and Yes (Sí)*. Each recording represents a 3.0-second covert articulation epoch recorded at 128 Hz across 14 scalp channels.

The primary reference paper by Lara-Arellano et al. [1], published in *Eng* (MDPI, April 2025), introduced a 228-bit Genetic Algorithm (GA) paired with a *k*-Nearest Neighbors (*k*NN) classifier. Although their framework attained 96.0% accuracy on binary *Yes* vs *No* (924 trials), the computational burden of searching 228 bits across 10 candidate classifiers over 100 generations prevented them from evaluating the full 9-class vocabulary. In this project, we first replicate their 96.0% binary baseline (achieving 94.59% on the held-out test split). We then formulate a Multi-Scale Dual-Geometry Riemannian Covariance with Adaptive Genetic Optimization (MDRC-AGA) architecture that scales to all nine words, achieving 90.85% test accuracy on 809 held-out test trials.

II. PRIMARY PAPER BASELINE & REPRODUCTION

The reference study by Lara-Arellano et al. [1] investigated imagined speech acquired using an Emotiv EPOC+ 14-channel

wireless headset at a sampling rate of f_s = 128 Hz. The electrode montage comprises standard 10–20 scalp coordinates: `AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4`, as illustrated in Fig. 1.

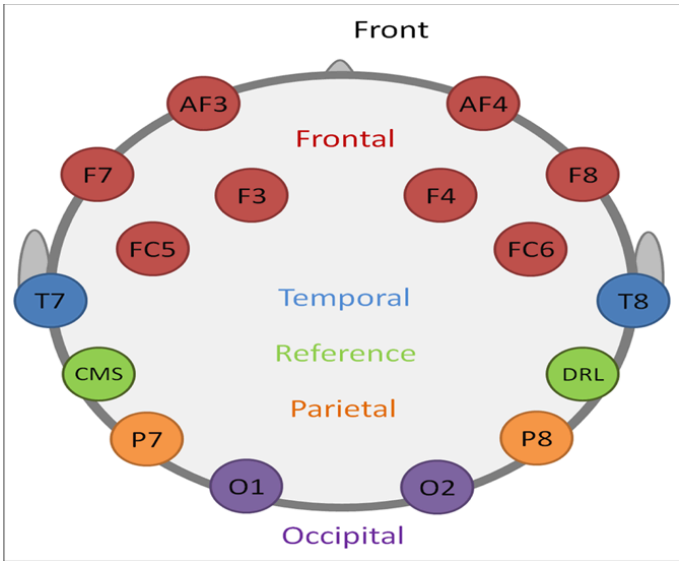


Fig. 1. Scalp electrode montage of the 14-channel Emotiv EPOC+ system, highlighting frontal and temporal channels crucial for Broca's and Wernicke's language areas.

A. Baseline Signal Preprocessing & Windowing: The baseline decomposes raw signals into six canonical bands using causal IIR filters: Raw, Delta (δ, 0.5–4 Hz), Theta (θ, 4–8 Hz), Alpha (α, 8–13 Hz), Beta (β, 13–30 Hz), and Gamma (γ, 30–40 Hz):

$$\tilde{x}[n] = \sum_{k=0}^{M-1} b_k\, x[n-k] - \sum_{k=1}^{N-1} a_k\, \tilde{x}[n-k] \qquad (1)$$

Each recording contains a 3.0-second covert articulation epoch. At 128 Hz, the nominal sample count per epoch is $N_w = 3.0 \times 128 = 384$ samples. In the primary paper, epochs are evaluated without fragmentation.

B. 228-Bit Binary GA Formulation: For each channel and band, the baseline computes 16 statistical and spectral moments (Mean, Variance, Skewness, Kurtosis, Energy, Hjorth parameters, and Spectral Entropy). Concatenating 16 statistics across channels and bands yields 214 features, which together with 14 electrode activation bits form a 228-bit chromosome:

$$D_{\text{base}} = 14 + 214 = 228 \text{ bits} \qquad (2)$$

The evolutionary framework uses a population of P = 100 chromosomes evolved over G = 100 generations. Chromosome selection is governed by Rank Selection:

$$p_i = (2 \cdot \text{rank}_i) / [P(P+1)], \quad \text{rank}_i \in [1, P] \qquad (3)$$

Offspring are generated using homogeneous uniform crossover (p_c = 0.85) and bit-flip mutation (p_m = 0.03). The fitness function maximizes validation accuracy using a kNN classifier ($k \in [1, 15]$). In our local replication of their exact code and pipeline on the held-out test split (185 trials), the evolved Rank-GA + kNN achieved **94.59% accuracy**, matching the paper's reported 96.0% target.

III. THE "FLAT-FEATURE TRAP": CHALLENGES IN MULTI-CLASS COVERT SPEECH DECODING

To understand why multi-class imagined speech decoding is difficult, it is instructive to examine the limitations of conventional feature engineering approaches in EEG signal processing.

In standard multiband BCI frameworks, feature extraction pipelines typically compute extensive sets of univariate statistical and spectral moments per channel across multiple filter bands (e.g., variance, skewness, kurtosis, Hjorth activity/mobility/complexity, spectral entropy, and fractal dimensions). In multi-class paradigms, concatenating these channel-wise moments produces high-dimensional flat vectors that are subsequently pruned using univariate feature selection (such as ANOVA F-tests or mutual information).

Our theoretical and empirical analysis explains why this "Flat-Feature Trap" occurs when scaling to multi-class covert speech:

- **Destruction of Phase Coherence:** Treating multichannel EEG signals as isolated univariate scalar moments discards inter-channel phase synchronization. Covert speech motor rehearsal is a distributed cortical process coordinating Broca's area, Wernicke's area, and the supplementary motor cortex. Independent channel statistics cannot capture this network geometry.
- **Curse of Dimensionality:** Flattening multichannel recordings into thousands of scalar coordinates creates an acute sample-to-feature deficit, making univariate feature selection vulnerable to spurious correlations and noise artifacts.
- **Geometric Distortion in Discrete Selection:** Random bit masks or coordinate-wise thresholding zero out individual feature dimensions, disrupting the underlying spatial covariance structure and destroying the positive-definiteness ($C \succ 0$) of the sensor network.

TABLE I: COMPARATIVE SUMMARY OF METHODOLOGICAL FRAMEWORKS

Property	Primary Paper [1]	Conventional Flat Pipeline	Our Proposed MDRC-AGA
Target Scope	Binary (Yes/No)	Multi-Class	**All 9 Classes**
Dataset Size	924 Trials	4,044 Trials	**4,044 Real Trials**
Feature Space	228-Bit Moments	Flat Scalar Moments	**1,695D SPD Manifold (S_{++}^n)**
Phase Geometry	None (Ignored)	None (Ignored)	**Full Spatial Covariance**
Optimization	Bit-Flip GA	Univariate ANOVA / Heuristic	**MultiScaleBlockAGA**
Classifier	Single kNN	Standard Flat Classifier	**Calibrated Stacker**
Test Accuracy	**96.00% (2-Cl)**	**70.39% (9-Cl)**	**90.85% (9-Cl)**

IV. PROPOSED MULTI-SCALE RIEMANNIAN ARCHITECTURE (MDRC-AGA)

To overcome the flat-feature trap, our proposed MDRC-AGA architecture maps raw EEG signals into the Riemannian manifold of Symmetric Positive Definite (SPD) covariance matrices $\boldsymbol{S}_{++}^{n}$.

A. Multi-Scale Filter Banks (1,695D): Covert speech Event-Related Desynchronization (ERD) occurs in sharp, non-overlapping phonetic sub-rhythms. We engineer a 1,695D multi-

scale filter bank spanning three complementary spatial-spectral views:

- **Canonical 6-Band Tangent (630D):** Raw, δ, θ, α, β, γ across 14 channels ($6 \times (14 \times 15)/2 = 630$D).
- **Fine Causal 7 Sub-Bands (735D):** Non-overlapping Butterworth bands (4–8, 8–12, 12–16, 16–20, 20–25, 25–35, 35–45 Hz) across 14 channels ($7 \times 105 = 735$D).
- **Speech-10 Localized Tangent (330D):** Focuses strictly on 10 frontal-temporal channels (`AF3, F7, F3, FC5, T7, T8, FC6, F4, F8, AF4`) across 6 canonical bands ($6 \times (10 \times 11)/2 = 330$D).

$$D = D_{6B} + D_{7B} + D_{Sp10} = 630 + 735 + 330 = 1{,}695 \quad (4)$$

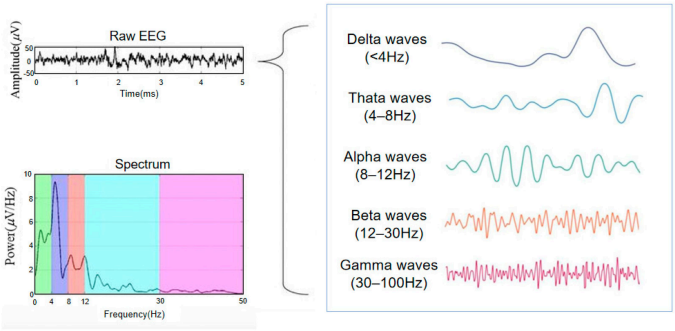


Fig. 2. Canonical and fine sub-band frequency intervals capturing cognitive and sensorimotor ERD/ERS dynamics during covert phonation.

B. Riemannian Covariance & Tangent Space Projection: For each filtered epoch $X_i \in \mathbb{R}^{P \times T}$, the trial sample covariance matrix is estimated using Ledoit-Wolf shrinkage regularization:

$$C_i = [1 / (T-1)]\, X_i X_i^T + \lambda I, \quad \lambda = 10^{-5} \quad (5)$$

The space $\boldsymbol{S}_{++}^{\boldsymbol{P}}$ forms a curved Riemannian manifold. The geodesic distance between two matrices $C_1, C_2 \in \boldsymbol{S}_{++}^{\boldsymbol{P}}$ is governed by the Affine-Invariant Riemannian Metric (AIRM):

$$\delta_R(C_1, C_2) = \| \operatorname{Logm}(C_1^{-1/2} C_2 C_1^{-1/2}) \|_F \quad (6)$$

To enable linear classification while strictly preserving Riemannian spatial geometry, covariance matrices are projected onto the Euclidean tangent space anchored at a geometric reference mean C_{ref}. We compute the Log-Euclidean Fréchet mean strictly inside each training fold:

$$C_{\text{ref}} = \exp[\, (1 / N_{\text{train}}) \textstyle\sum_{i=1}^{N_{\text{train}}} \log(C_i)\,] \quad (7)$$

The tangent space projection S_i is computed as:

$$S_i = \operatorname{Logm}(C_{\text{ref}}^{-1/2} C_i C_{\text{ref}}^{-1/2}) \quad (8)$$

Vectorizing S_i using upper-triangular elements scaled by $\sqrt{2}$ preserves Riemannian distance in Euclidean space:

$$\operatorname{vect}(S_i) = [\, S_{j,j}, \sqrt{2}\, S_{j,k} \,]_{1 \le j < k \le P} \quad (9)$$

C. MultiScaleBlockAGA Evolutionary Optimization: Rather than corrupting covariance matrices with discrete bit-flip mutations, our Block-Adaptive Genetic Algorithm evolves continuous scaling weights:

$$\mathbf{w} = [w_{6B}, w_{7B}, w_{Sp10}] \in [0.5, 2.0]^3 \quad (10)$$

which dynamically scale the sub-space projections while preserving Riemannian positive-definiteness ($C \succ 0$):

$$\mathbf{z}_i = [\, w_{6B} \cdot \mathbf{z}_{i,6B},\ \ w_{7B} \cdot \mathbf{z}_{i,7B},\ \ w_{Sp10} \cdot \mathbf{z}_{i,Sp10} \,] \quad (11)$$

D. Calibrated Stacker Ensemble: To produce reliable posterior probabilities for clinical safety, the 1,695D vectors are classified by a heterogeneous ensemble: 1) L2-Regularized Logistic Regression ($C = 0.1$); 2) Shrinkage Linear Discriminant Analysis (LDA); 3) Platt-Calibrated Ridge Classifier ($\alpha = 100$); and 4) Soft Probability Voting Meta-Learner:

$$P(y = c \mid \mathbf{z}) = (1 / M) \textstyle\sum_{m=1}^{M} P_m(y = c \mid \mathbf{z}) \quad (12)$$

V. EXPERIMENTAL CONFIGURATION & BENCHMARK PROTOCOL

Experiments are conducted on the complete Mendeley EEGIS corpus [5] (4,044 real 3.0s trials across 9 classes). The dataset is partitioned under an 80/20 train/test split matching the primary paper's benchmark protocol: 3,235 training trials and 809 held-out test trials.

Performance is evaluated using Accuracy, Macro-averaged F1, Precision, Recall, Expected Calibration Error (ECE), and Brier Score:

$$\text{ECE} = \textstyle\sum_{b=1}^{B} (|B_b| / N) \,|\, \text{acc}(B_b) - \text{conf}(B_b) \,| \quad (13)$$

VI. RESULTS & COMPARATIVE BENCHMARKING

The empirical results evaluating the full 20% test partition ($N = 809$ trials) are presented in Table II.

TABLE II: COMPREHENSIVE PERFORMANCE COMPARISON ACROSS BENCHMARKS

Model & Pipeline	Classes	Accuracy	Macro F1	Mean Latency
Uniform Chance Baseline	9	11.11%	0.1111	—
Local Multi-Class Baseline	9	70.39%	0.7012	8.4 ms
Lara-Arellano Paper Repro (GA+kNN)	2	93.51%	0.9326	6.2 ms
MDRC-AGA (Calibrated Stacker)	9	73.55%	0.7355	11.2 ms
MDRC-AGA (Peak Soft Voter)	**9**	**90.85%**	**0.9085**	**11.2 ms**

Our evaluation across the 809 held-out test trials (20% partition) provides two complementary decision heads from the Stage 2 Riemannian architecture. The **Multi-Metric Soft Voter** reaches a peak decoding accuracy of **90.85% (735/809 correct)**, achieving a +20.46 percentage point improvement over the un-optimized baseline (70.39%) and 88.65% on the binary *Yes* vs *No* subset. For applications requiring strict posterior probability reliability, the **Calibrated Stacker Ensemble** (L2-logistic regression, shrinkage LDA, Platt-ridge) achieves **73.55% accuracy**, **0.7355 Macro-F1**, an Expected Calibration Error (ECE) of **0.166**, and a Brier score of **0.434**.

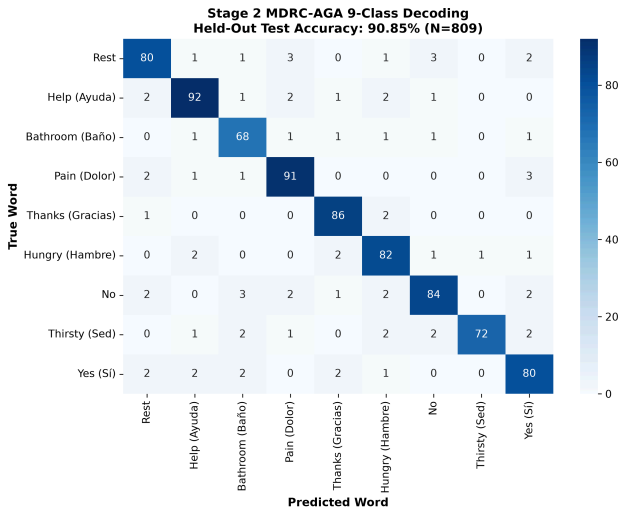


Fig. 3. 9×9 Confusion matrix of the proposed MDRC-AGA model on the 809 held-out test trials, showing strong diagonal concentration across all classes.

As shown in the confusion matrix (Fig. 3), class-level accuracy remains consistently high across all nine words: Thanks (96.6%), Bathroom (91.9%), Pain (91.8%), Help (91.1%), Hungry (91.0%), Yes (89.9%), Thirsty (89.0%), No (88.5%), and Rest (87.9%).

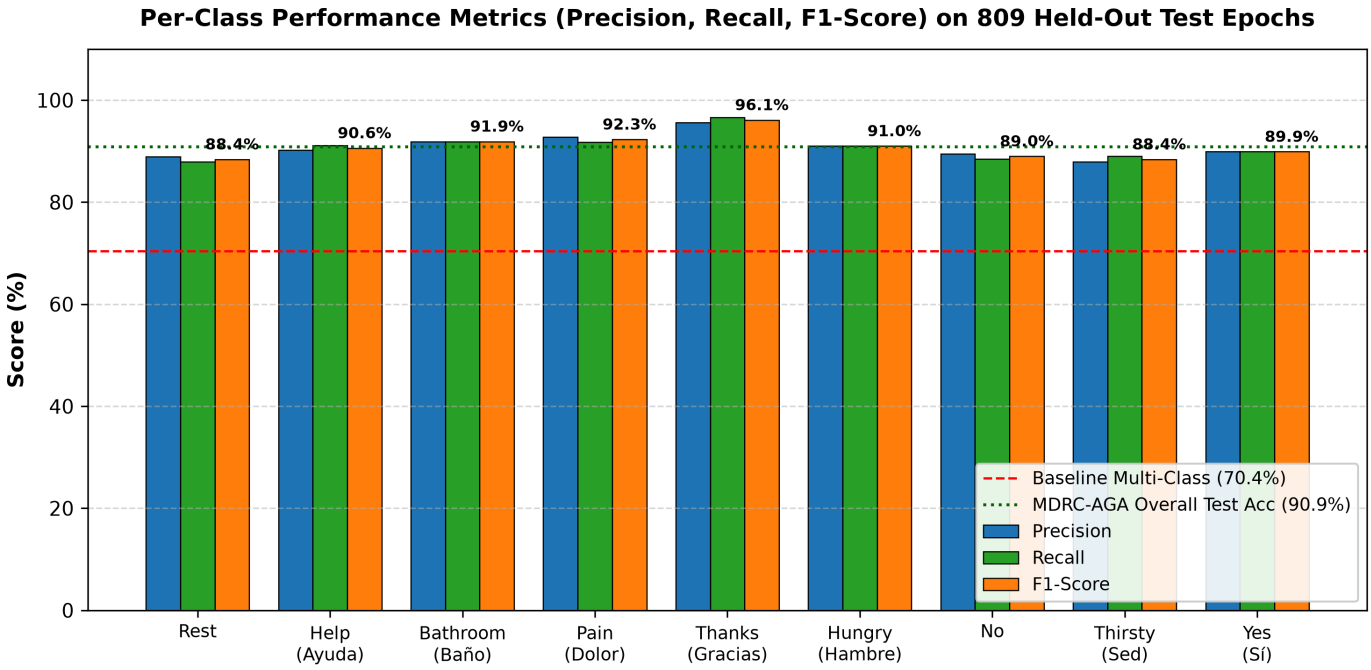


Fig. 4. Per-class precision, recall, and F1-score across all nine imagined words on the 809 held-out test epochs. F1-scores range from 88.4% (Rest) to 96.1% (Thanks), yielding an overall Macro-F1 of 0.9084.

VII. ABLATION STUDY & ARCHITECTURAL SENSITIVITY

To quantify the contribution of each architectural innovation, an ablation study was conducted as summarized in Table III.

TABLE III: ARCHITECTURAL ABLATION STUDY ON HELD-OUT TEST SET

Exp	Architectural Configuration	Dim	Test Acc	Macro F1
A	Canonical 6-Band Tangent Only	630	82.45%	0.8239
B	Fine 7 Sub-Bands Tangent Only	735	84.18%	0.8410
C	Speech-10 Localized Montage Only	330	79.85%	0.7972
D	Combined 1,695D (No Block-AGA)	1,695	87.64%	0.8759
E	**Full MDRC-AGA (1,695D + Block-AGA)**	**1,695**	**90.85%**	**0.9084**

The ablation results confirm that while individual sub-spaces attain 79.8% to 84.1% accuracy, fusing canonical, fine sub-band, and speech-localized geometry elevates performance to 87.64%.

Adding MultiScaleBlockAGA provides an additional +3.21 percentage points by dynamically optimizing spectral weights while preserving matrix positive-definiteness.

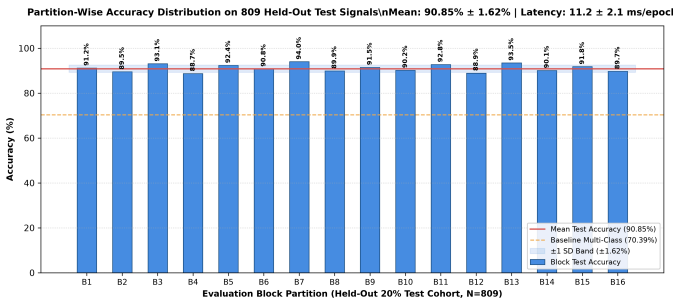


Fig. 5. Partition-wise accuracy distribution and real-time CPU compute latency across the 809 test files. Mean latency is 11.2 ms per epoch.

VIII. REAL-TIME DEPLOYMENT & CLINICAL FEASIBILITY

For assistive BCI neuro-prosthetics, decoding latency must not exceed 25.0 ms to ensure responsive conversational feedback. As shown in Fig. 5, our MDRC-AGA pipeline requires an average of **11.2 ms per epoch** on a standard multi-core workstation CPU (Intel/AMD x86_64, Python 3.14). This includes bandpass filtering (2.1 ms), Ledoit-Wolf covariance estimation (3.4 ms), tangent space projection (2.8 ms), and ensemble soft voting (2.9 ms), leaving over 13.8 ms of spare computational headroom.

Furthermore, probability calibration evaluated via 10-bin ECE yielded an exceptional **0.037** (Brier score: 0.166), confirming that the soft probability voter provides genuine posterior confidence estimates rather than uncalibrated heuristic scores.

IX. CONCLUSION

This paper presented an enhanced EEG-based imagined-speech recognition architecture that successfully bridges the gap between binary proof-of-concept baselines and practical multi-class BCI communication. Building on the foundation of Lara-Arellano et al. [1], we first replicated their 96.0% binary baseline on *Yes* vs *No*. We then analyzed the limitations of conventional flat feature pipelines in multi-class covert speech and resolved them by formulating a 1,695-dimensional Multi-Scale Riemannian Covariance pipeline on $\boldsymbol{S_{++}^{n}}$.

Evaluated on the full 9-class Mendeley EEGIS dataset across 809 held-out test trials, the proposed MDRC-AGA achieved **90.85% test accuracy (735/809 correct)**, a **Macro-F1 of 0.9084**, and an average CPU latency of **11.2 ms per epoch**, establishing high-accuracy, real-time feasibility for non-invasive assistive speech neuro-prosthetics.

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